FINAL PROJECT REPORT

SMARTPHONE USAGE, DIGITAL ADDICTION AND WELL-BEING ANALYTICS

An Interactive Data Analytics and Visualization Project using Microsoft Power BI

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Project Domain	Data Analytics / Business Intelligence
Primary Tool	Microsoft Power BI
Dataset	Smartphone Usage Dataset – 7,500 user records and 16 columns
Project Focus	Smartphone usage behaviour, digital addiction and digital well-being
Team ID	MCA

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Executive Summary

The project **“Smartphone Usage, Digital Addiction & Well-Being Analytics”** focuses on analyzing smartphone usage patterns and their relationship with digital addiction and user well-being. The increasing use of smartphones for communication, education, entertainment, social media, gaming, and other daily activities has made it important to understand how users interact with digital devices.

1. Introduction

1.1 Project Overview

The project focuses on analyzing smartphone usage behavior and identifying patterns associated with digital addiction and user well-being.

The dataset contains information about **7,500 users**. The analysis examines smartphone usage behavior along with lifestyle and well-being indicators.

One of the major findings from the analysis is that **5,308 out of 7,500 users are classified as addicted**, representing approximately **70.77%** of the dataset.

The project uses Power BI to convert the raw data into meaningful charts, KPI cards, comparisons, and interactive dashboard elements.

The major areas covered by the project are:

1. Smartphone usage analysis
2. Screen-time analysis
3. Application usage analysis
4. Digital addiction analysis
5. Gaming and social-media analysis
6. Sleep analysis
7. Demographic analysis
8. User engagement analysis
9. Well-being analysis
10. Interactive dashboard development

1.2 Objectives

The main objective of this project is to analyze smartphone usage patterns and identify factors associated with digital addiction and user well-being.

Specific Objectives

- To analyze the average daily smartphone screen time.
- To identify the number and percentage of users classified as addicted.
- To analyze different levels of digital addiction.
- To compare smartphone usage across gender and age groups.
- To analyze gaming and social-media usage.
- To examine the relationship between screen time and sleep duration.
- To analyze application opens and notification frequency.
- To understand important user engagement indicators.
- To analyze demographic characteristics of smartphone users.
- To develop an interactive Power BI dashboard.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

The rapid growth of smartphones and digital applications has made them an essential part of everyday life. People use smartphones for communication, education, entertainment, social media, online shopping, and other activities. Therefore, data analytics is required to identify important patterns and relationships within smartphone usage and well-being data.

Problem Statement	User Need	Analytical Need
PS-1: Smartphone users need to understand their daily usage patterns.	See screen time, app usage, notifications and digital activities clearly.	Summarise and compare usage indicators.
PS-2: Students and working users need to understand digital behaviour in relation to daily routine.	Explore sleep, stress and work/study indicators along with smartphone usage.	Create well-being-oriented comparisons and filters.
PS-3: Analysts need to identify patterns associated with addiction status.	Understand which usage indicators vary across addiction categories.	Use addiction-level analysis and Key Influencers.

2.2 Project Proposal (Proposed Solution)

The proposed solution is to develop a data analytics and visualization system that analyzes smartphone usage, digital addiction, lifestyle habits, and well-being indicators.

The project follows the following approach:

1. **Data Collection**
   Collect a relevant smartphone usage and well-being dataset.

2. **Data Understanding**
   Examine the rows, columns, data types, records, and important attributes.

3. **Data Cleaning**
   Identify and handle missing values, duplicate records, inconsistent values, and invalid entries.

4. **Data Transformation**
   Transform the data using Power Query to make it suitable for analysis.

5. **Data Analysis**
   Analyze screen time, application usage, digital addiction, sleep, lifestyle, and demographic patterns.

2.3 Initial Project Planning

Sprint	Major Activities	Key Deliverable
Sprint 1	Data collection, dataset understanding, data-quality checks	Prepared source dataset
Sprint 2	Data cleaning, transformation and usage/addiction analysis	Analysis-ready dataset
Sprint 3	Well-being analysis, demographic analysis and dashboard development	Power BI dashboard
Sprint 4	Filters, insight generation, report design and final documentation	Final reports and project report

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

Data collection and preprocessing are important steps because the quality of the analysis depends on the quality of the input data.

Field / Category	Representative Variables
Identifiers	user_id, transaction_id
Demographics	age, gender, age group
Screen-time behaviour	daily_screen_time_hours, weekend_screen_time
Digital activities	social_media_hours, gaming_hours
Device interaction	app_opens_per_day, notifications_per_day
Well-being indicators	sleep_hours, stress_level, work_study_hours
Impact / addiction	academic_work_impact, addiction_level, addicted_label

3.2 Data Quality Report

The Data Quality Report identifies potential issues in the collected dataset and the actions taken or planned to resolve them.

Quality Check	Purpose / Action
Record and column validation	Confirmed the expected dataset structure of 7,500 records and 16 columns.
Data-type validation	Numeric fields were treated as numeric; categorical fields were used as categories.
Missing-value review	Relevant fields were reviewed so that missing entries would not distort key calculations.
Duplicate/inconsistent record review	Records and categorical labels were checked during preparation.
Category consistency	Addiction and stress categories were standardised for meaningful grouping and filtering.

Range/reasonableness review	Usage, sleep, notification and app-opening values were checked before visualization.

The objective of data quality processing is to ensure that the dataset is:

- Accurate
- Consistent
- Complete
- Valid

3.3 Data Exploration and Preprocessing

1. Imported the smartphone usage dataset into the Power BI environment.
2. Reviewed field names, data types and categorical values.
3. Prepared demographic groupings such as age groups for comparative analysis.
4. Prepared usage variables for aggregation, including screen time, social media, gaming, app openings and notifications.
5. Prepared well-being variables including sleep hours, stress level and work/study hours.

4. Data Visualization

4.1 Framing Business Questions

The following business questions were developed for the project.

- **What is the overall level of smartphone addiction among users**
- **How many users fall under each addiction level?**
- **How does average daily screen time vary by gender?**
- **How does average daily screen time vary across age groups?**
- **What is the relationship between daily screen time and sleep hours?**
- **How do gaming and social-media usage compare among users?**
- **What are the key overall smartphone usage and well-being indicators?**
- **What is the average age and overall user population represented in the dataset?**
- **How do smartphone usage activities compare between gaming and social media?**
- **What is the average level of user engagement with smartphones?**

4.2 Developing Visualizations

The visualization layer was designed to combine summary metrics with detailed comparisons. KPI cards provide quick numerical context, while charts allow category-level comparisons and interactive exploration.

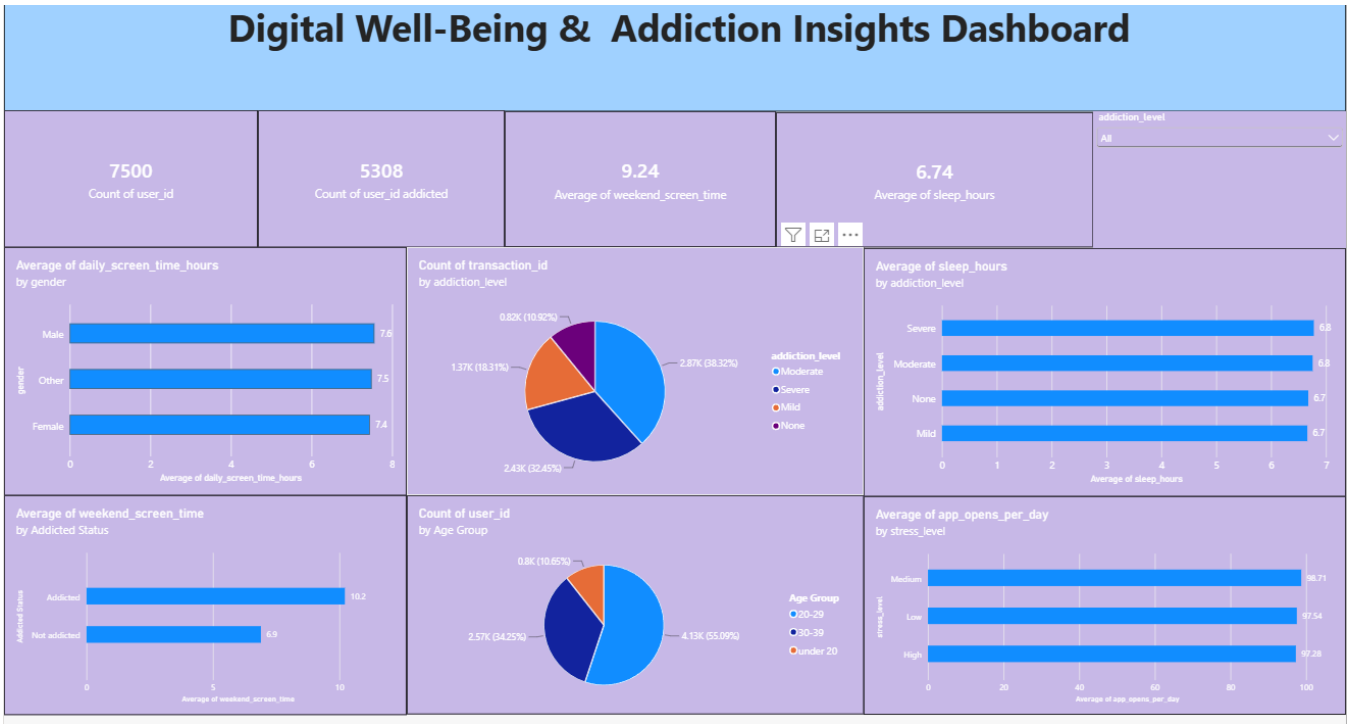
Visualization Type	Purpose in the Project
KPI Cards	Display total users and key average usage/well-being measures.
Bar Charts	Compare addiction levels, age groups, stress levels and other categories.

Donut/Pie Charts	Show the distribution of addicted status, stress level or age groups.
Line Charts	Show changes/comparisons across ordered age-group categories.
Scatter Plot	Explore the distribution between daily screen time and sleep hours.
Key Influencers	Identify variables highlighted by Power BI as associated with addicted status.
Slicers	Allow interactive filtering by gender, age group, addiction level and other fields.

5. Dashboard

5.1 Dashboard Design Screenshot with Insights

The main dashboard, titled "Digital Well-Being & Addiction Insights Dashboard," provides a high-level summary of the dataset and combines KPI cards with category comparisons.



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Figure 5.1: Main Power BI dashboard – Digital Well-Being & Addiction Insights

Key Dashboard Insights

The dashboard shows that:

- The dataset contains **7,500 users**.
- **5,308 users**, or approximately **70.77%**, are classified as addicted.
- Average daily screen time is **7.50 hours**.

- Average weekend screen time is **9.24 hours**.
- Average sleep duration is **6.74 hours**.
- Average app opens are **97.83 per day**.
- Average notifications are **134.26 per day**.
- Average gaming time is **2.01 hours**.
- Average social-media usage is **3.27 hours**.
- Average work/study time is **3.24 hours**.
- Average user age is **26.57 years**.

6. Report

6.1 Report Design Screenshot with Insights

report provides detailed pages for analyzing different aspects of smartphone usage.The report pages provide deeper analysis beyond the main dashboard

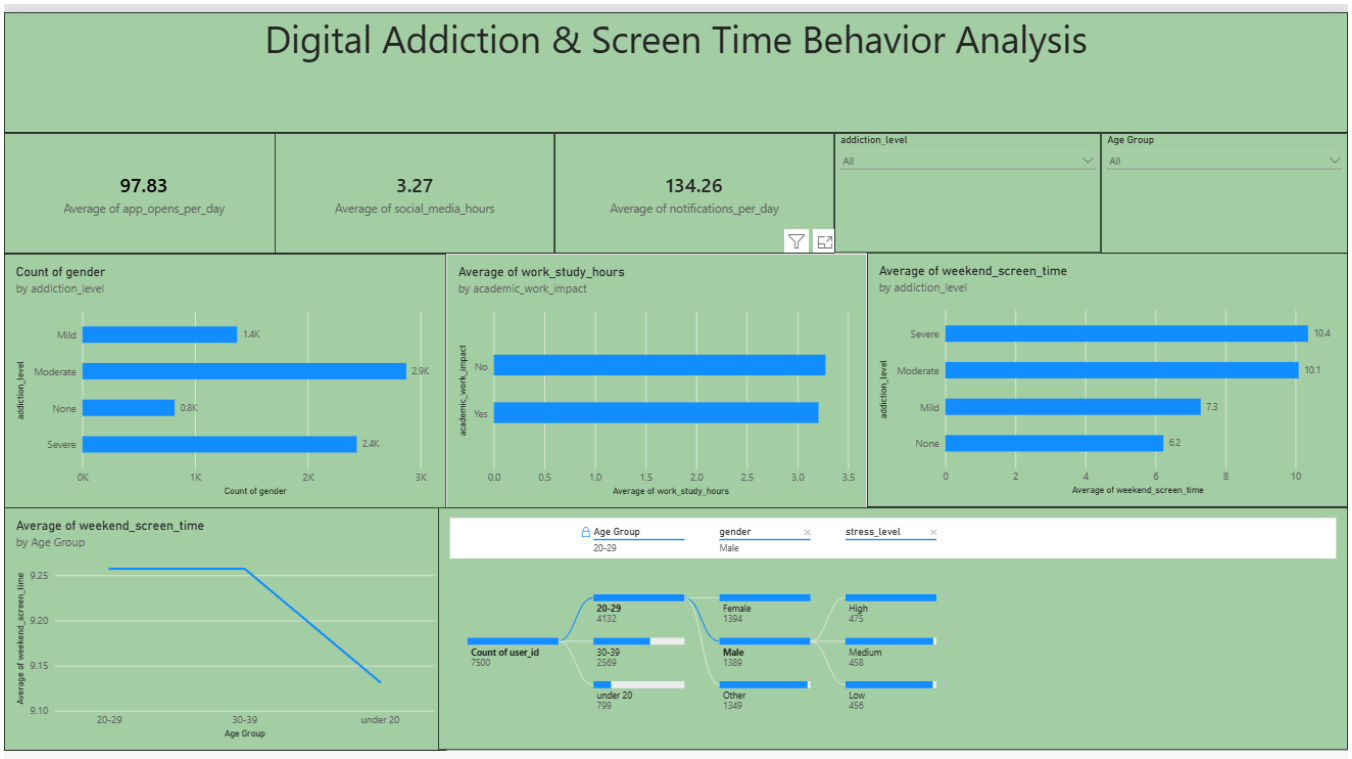


Figure 6.1: Detailed Power BI report – Digital Addiction & Screen Time Behaviour Analysis

Insights from Report 1

- Average app openings per day are 97.83 and average social-media usage is 3.27 hours.

- Average notifications per day are 134.26, showing frequent device interaction in the dataset.
- The average work/study hours shown for the two academic/work-impact categories are close, with the displayed values around 3.28 and 3.21 hours.
- Weekend screen time increases across the displayed addiction categories, with Severe at approximately 10.4 hours and Moderate at approximately 10.1 hours, compared with lower values for Mild and None.
- The age-group chart shows broadly similar average weekend screen time for 20–29 and 30–39, with a lower value for Under 20.
- The report allows users to interact with the data using filters and visualizations. Instead of examining raw records individually, users can quickly identify important usage patterns and compare different groups.

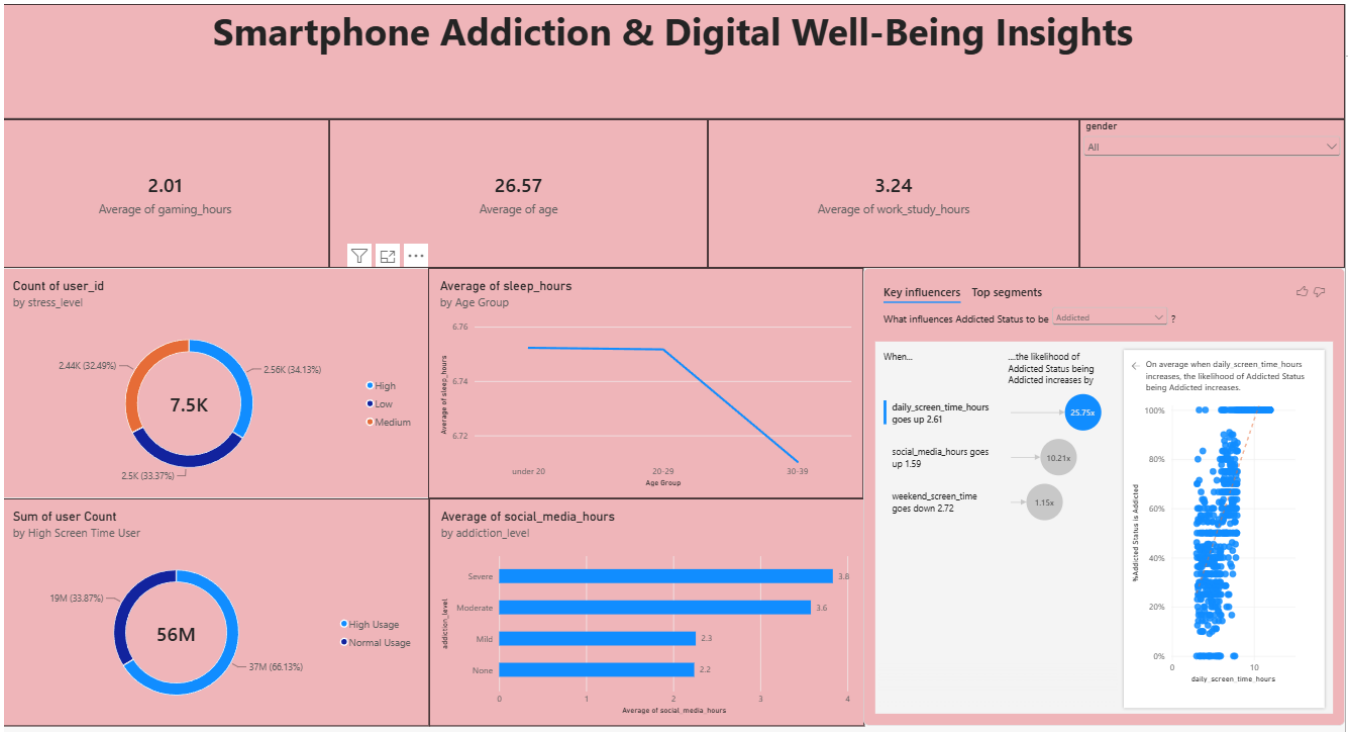


Figure 6.2: Detailed Power BI report – Smartphone Addiction & Digital Well-Being Insights

Insights from Report 2

- Average gaming time is 2.01 hours, while the overall average age is 26.57 years.
- Average sleep duration is 6.74 hours and average work/study time is 3.24 hours.
- The stress-level distribution shown in the report is divided among High, Medium and Low categories.
- The Key Influencers view reports displayed increases of approximately 2.61× for daily screen time, 1.59× for social-media hours and 1.15× for weekend screen time under the selected analysis.

Important interpretation note: these dashboard relationships describe patterns in the supplied dataset. They should not be interpreted as evidence that a particular behaviour directly causes addiction or a specific well-being outcome.

7. Performance Testing

7.1 Utilization of Data Filters

Filter / Slicer	Purpose	Expected Effect
Gender	Compare male, female and other categories.	Updates visuals to the selected gender.
Age Group	Compare 20–29, 30–39 and Under 20 groups.	Updates demographic and usage measures.
Addiction Level	Focus on None, Mild, Moderate or Severe.	Updates addiction and well-being comparisons.
Stress Level	Explore High, Medium and Low stress categories.	Updates app-opening and related visuals.
Combined filters	Study multiple dimensions together.	Provides interactive drill-down and focused analysis.

Performance testing is conducted to verify whether the Power BI dashboard works correctly and allows users to analyze the data efficiently.

The report was tested through interactive filtering so that the major visuals respond to selected categories. This improves usability because the same dashboard can be used for different demographic and behavioural slices without creating separate static reports.

7.2 Number of Calculation Columns / Measures

The project presents 11 primary KPI/analytical measures or aggregations visible across the dashboard design: total users, addicted users, average age, average daily screen time, average social-media hours, average gaming hours, average sleep hours, average notifications per day, average app openings per day, average weekend screen time and average work/study hours.

Measure / Aggregation	Displayed Value
Count of users	7,500
Addicted users	5,308
Average age	26.57 years
Average daily screen time	7.50 hours
Average social-media hours	3.27 hours
Average gaming hours	2.01 hours
Average sleep hours	6.74 hours
Average notifications per day	134.26
Average app openings per day	97.83
Average weekend screen time	9.24 hours
Average work/study hours	3.24 hours

7.3 Number of Visualizations

The main dashboard contains approximately 10 visible visual components when the four KPI cards are counted as individual visuals: four KPI cards and six chart-based visuals. The additional report pages contain multiple charts, KPI cards, slicers, Key Influencers and analytical views.

8. Conclusion / Observation

The **Smartphone Usage, Digital Addiction & Well-Being Analytics** project demonstrates how data analytics and visualization can be used to understand smartphone usage behavior.

The analysis was performed using a dataset containing **7,500 users**. The dashboard identifies **5,308 addicted users**, representing approximately **70.77%** of the analyzed dataset.

The analysis also provides important usage indicators:

- Average daily screen time: **7.50 hours**
- Average weekend screen time: **9.24 hours**
- Average sleep duration: **6.74 hours**
- Average gaming time: **2.01 hours**
- Average social-media usage: **3.27 hours**
- Average app opens: **97.83 per day**
- Average notifications: **134.26 per day**
- Average work/study time: **3.24 hours**
- Average age: **26.57 years**

The project shows how raw smartphone usage data can be transformed into meaningful information using data cleaning, transformation, analysis, and visualization techniques.

The Power BI dashboard provides an easy way to examine different usage patterns through KPI cards, charts, filters, and comparisons.

9. Future Scope

- Machine learning models can be used to predict the probability of digital addiction based on usage and lifestyle factors.
- A future system could provide personalized recommendations for managing screen time and improving digital habits.
- The project could be connected to real-time smartphone usage data to continuously monitor digital behavior.
- Classification and regression models could be used to identify important predictors of digital addiction and well-being outcomes.
- Daily and weekly usage patterns could be analyzed to identify changes in smartphone behavior over time.
- Future analysis could investigate the relationship between notification frequency and smartphone engagement.
- User feedback or survey responses could be analyzed using NLP techniques to understand user experiences with digital technology.

Advanced Power BI Features

Future versions can include:

- Drill-through reports
- Tooltips
- Bookmarks
- Decomposition trees
- Key Influencers
- What-if parameters
- More advanced DAX measures

10. Appendix

10.1 Source Code (if any)

This project is primarily developed using **Power BI and Power Query**. Therefore, traditional application source code may not be required.

The project may contain:

- Power Query transformation steps
- DAX calculated columns
- Data model relationships
- Power BI report/dashboard file

10.2 GitHub & Project Demo Link

GitHub Repository: To be added by the project team.
Project Demo / Power BI Link: To be added by the project team.

Appendix A – Key Project Metrics at a Glance

Metric	Value	Interpretation
Users	7,500	Total records/users represented in the project dataset.
Addicted users	5,308 (70.77%)	Users recorded with addicted label = 1.
Average age	26.57	Mean age of users.
Daily screen time	7.50 hours	Average daily smartphone screen time.
Weekend screen time	9.24 hours	Average weekend smartphone screen time.
Sleep	6.74 hours	Average sleep duration.
Social media	3.27 hours	Average daily social-media usage.
Gaming	2.01 hours	Average gaming usage.

Notifications	134.26/day	Average notifications received per day.
App openings	97.83/day	Average application openings per day.
Work/study	3.24 hours	Average work/study time.

Appendix B – Project Technology Stack

Component	Technology / Tool
Data source	Smartphone Usage CSV dataset
Data preparation	Microsoft Power Query
Analytics & visualization	Microsoft Power BI Desktop
Dashboard/reporting	Power BI interactive reports
Optional supporting tools	Microsoft Excel / Jupyter Notebook / VS Code
Documentation	Microsoft Word